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Exercise Sheet 1: Logical Operators

We define the binary logical operator $\star$ as follows via its truth table:

A	B	$A \star B$
0	0	1
0	1	1
1	0	1
1	1	0

Task 1

(5 Points)

Prove or disprove that the operator $\star$ is associative:

$$(A \star B) \star C \equiv A \star (B \star C)$$

by computing the truth tables for the left-hand side and right-hand side and comparing them.

Task 2

(5 Points)

Prove that $A \star A$ is equivalent to the negation of A :

$$A \star A \equiv \neg A$$

by computing the truth tables for both sides and comparing them.

Task 3

(10 Points)

Given the following truth table for a function F :

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Find a formula G that uses only the logical operator $\star$ and contains each of the propositional symbols A , B , and C at least once, such that $G \equiv F$. No justification is required.

Task 4

(8 Points)

Construct the complete truth table for the compound proposition $(A \wedge B) \rightarrow (\neg C \vee A)$.

Your truth table should show the values of all intermediate expressions:

- $A \wedge B$
- $\neg C$
- $\neg C \vee A$
- $(A \wedge B) \rightarrow (\neg C \vee A)$

Remember that $P \rightarrow Q$ is equivalent to $\neg P \vee Q$, so it's only false when P is true and Q is false.

Task 5

(7 Points)

Prove or disprove the following logical equivalence using truth tables:

$$\neg(A \wedge B) \equiv \neg A \vee \neg B$$

This is one of De Morgan's laws. Construct truth tables for both sides and compare them row by row. If they match in every row, the equivalence holds; if they differ in any row, provide a counterexample.